

Antisense RNAs (asRNAs) as key players in gallbladder cancer progression: a bioinformatics analysis

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ABSTRACT

Aim: This study reports differential expression of Antisense RNAs (asRNAs) by analyzing transcriptomic profiles in gallbladder cancer (GBC).

Background: asRNAs play crucial roles in developing various tumors. However, the presence and biological mechanism of asRNAs in GBC development are still unknown.

Methods: Differentially expressed asRNAs (DE-asRNAs) were systematically identified from RNA sequencing data \ from ten GBC patients. Functional enrichment analysis was performed, followed by the identification of mRNAs targeted by asRNAs and the construction of a gene regulatory network of asRNAs targeting mRNAs.

Results: Of the 891 asRNAs identified, 17 DE-asRNAs were statistically significant. Out of 17, 12 asRNAs were upregulated, and five asRNAs were downregulated. Functional enrichment analysis showed their role in methylation and developmental processes. Of the 17 asRNAs, 14 are novel (UNC5B-AS1, SLC2A1-AS1, BBOX1-AS1, SOX21-AS1, ELFN1-AS1, TRPM2-AS, DNAH17-AS1, DCST1-AS1, VPS9D1-AS1, MIR1-1HG-AS1, HAND2-AS1, PGM5P4-AS1, PGM5P3-AS1, and MAGI2-AS). Enrichment analysis of asRNAs with target mRNAs showed enrichment in biological regulation and developmental processes involved in the PI3K, p53, apoptosis, and VEGF signaling pathways.

Conclusion: This study identified 14 asRNAs for the first time and showed that asRNAs targeting mRNAs strongly associated with tumor development in GBC through the PI3KCA and TP53 pathways.

Keywords: Antisense RNA (asRNAs), Gene Ontology (GO), Pathway analysis, Disease Ontology (DO), asRNAs targeting mRNAs, Gene regulatory network.

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Introduction

Globally, gallbladder cancer (GBC) is the fifth-leading cause of gastrointestinal malignancy (1) and is a highly aggressive biliary tract neoplasm (2). The incidence of GBC shows marked geographical variations, with areas of high incidence such as Chile, Poland, and India (3). Developments in the identification and treatment of GBC have led to a stage shift, but the 5-year survival rate remains below 5% (4). There has been significant progress in understanding the genetic

processes underlying GBC; its molecular etiology is still not completely understood, and no definite pathways and driver genes have been identified. Thus, it is crucial to find novel targets for therapy and early screening biomarkers, and a clear understanding of the molecular pathways of carcinogenesis.

Noncoding ribonucleic acids (ncRNAs) have been recognized as significant regulators of tumor growth and metastasis (5). Antisense RNAs (asRNAs), identified by Simons et al. in 1983, are short, noncoding RNA molecules ranging from 19 to 23 nucleotides that bind to complementary mRNA sequences (6, 7). Antisense RNAs regulate gene expression at many stages, including replication, transcription, and translation (8). The application of

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asRNA is thought to replace conventional approaches for gene-specific silencing. Hence, asRNA is also known as a "yard waste" of RNA because of its crucial role in intracellular gene regulation (9).

The evidence suggests that asRNAs are associated with tumor development. They can potentially be both tumor drivers and tumor suppressors (10). Additionally, antisense RNAs can function as cancer-promoting (miR-21, miR-155, miR-17-20) and cancer-suppressing (miR-15, miR-16, miR-143) genes. WD40-encoding RNA antisense to p53 (WRAP53) (190) (11), HIF1A antisense RNA 2 (HIF1A-AS2) (201) (12), MALAT1 antisense RNA (TALAM1) (221) (13), and HOXA11 antisense RNA (HOXA11-AS) (206) (14) are some of the asRNAs that have shown a therapeutic role in cancer.

Earlier studies suggested that AFAP1-AS1 (actin filament-associated protein one antisense RNA 1) is linked to several malignancies (15), including GBC (16). Additionally, HMGA2-AS1 (high mobility group A asRNA) controls the translation of its sense gene and may act as a tumorigenic driver (17). The ERK/MAPK pathway has been shown to be induced by MALAT1 (metastasis-associated lung cancer transcript 1), which leads to the development and metastasis of GBC cells (18). A previous study reported that PGM5P4-AS1 could be an anticancer marker in several malignancies (19).

The interactions between asRNAs and other RNAs have helped us understand how asRNAs work in tumor development and progression (20). However, studies concerning GBC and biliary tract tumors are sparse. The present study aimed to analyze a transcriptome profile comprehensively, screen differentially expressed asRNAs (DE-asRNAs), and explore their role in gallbladder cancer progression using an in silico approach. In addition, the GBC asRNA-targeted mRNA regulatory network was built between the DE-asRNAs.

Methods

Patient tissue sample

Gallbladder cancer tissue samples (n=10) were obtained in RNA later (Thermo Fisher Scientific, Inc., Waltham, MA, USA) from patients undergoing surgery or biopsy. They were snap-frozen in liquid nitrogen and later stored at -80°C until further analysis after

obtaining written informed consent. The study was approved by the Institutional Ethics Committee.

RNA sequencing and data analysis

RNA isolation, cDNA library preparation, and sequencing were carried out following the procedure described by Dixit et al. (21). RNA was extracted from tissue using the TRIzol reagent (Thermo Fisher Scientific). The quality and quantity of extracted RNA were evaluated with an Agilent 2100 Bioanalyzer. cDNA synthesis was performed using the NEBNext® RNA First Strand Synthesis Module and the NEBNext® Ultra™ II Non-Directional RNA Second Strand Synthesis Module. The synthesized cDNA was then purified using SPRIselect Beads or NEBNext Sample Purification Beads. Finally, the prepared libraries were quantified using the DNA 1000 Chip on an Agilent Bioanalyzer.

The cuffdiff was used to compute the normalization and p-value. The test statistic used by cuffdiff were $T = E[\log(y)] / \text{Var}[\log(y)]$, where y is the ratio of the normalized counts between two conditions, and this ratio approximately follows a normal distribution. Hence, a t-test was used to calculate the p-value for differential expression. The normalization process used a geometric mean-based scaling method with at least three biological replicates per condition. Outliers were removed using the MAD method, missing data were imputed, and FPKM filtering was applied to exclude low-abundance transcripts. Differentially expressed genes (DEGs) were identified using a threshold of absolute log fold change (FC) > 3.0 and a significance level of $P < 0.05$. An adjusted P-value < 0.05 was set as the cut-off criterion.

Gene Ontology (GO) and functional enrichment analysis

Functional enrichment analysis was carried out to illustrate the biological functions and pathways of the DE-asRNAs.

Lantern (<https://sysbio.sitehost.iu.edu/lantern/>), a genome-scale archive for long noncoding RNA, was used to perform ontology annotation. ENCORI (The Encyclopedia of RNA Interactomes) (<https://starbase.sysu.edu.cn/>) was used to predict the associated pathways. The RNA target pathway analysis included only interactions validated in at least one experimental setup by setting the interaction and experiment number filters to ≥ 1 and followed by the selection of mammal in clade and the

human in genome (hg38). Statistical analysis was performed using a hypergeometric test to determine which pathways were significantly enriched. To reduce false positives and account for multiple hypothesis testing, FDR correction using the Benjamini-Hochberg method was applied, considering results significant if $FDR \leq 0.05$. The key biological pathways linked to differentially expressed antisense RNAs (DE-asRNAs) were identified by annotating them using ENCORI's integrated databases, which included KEGG, Reactome, and WikiPathways.

Disease Ontology (DO)

To explore disease associations for DE-asRNAs, MalaCards (<https://www.malacards.org>), which compiles disease-related information from 77 sources, including GeneCards, OMIM, ClinVar, Orphanet, and DisGeNET, and integrates these with ICD-11, was used. The official gene symbols corresponding to the DE-asRNAs were entered into the MalaCards search bar, leveraging its support for multiple identifiers such as aliases and Ensembl IDs to ensure accurate retrieval of gene-disease associations.

Screening, Identification of asRNAs targeted mRNAs and functional analysis

The RNA-associated interaction database (RNAInter - RNA Interactome Database) (<http://www.rnainter.org/>) was used to investigate

asRNAs targeted mRNAs. The RNA-RNA interaction in *Homo sapiens* with a confidence score of 0.0 to 1.0 was analyzed. In addition, the detection methods included those with strong experimental evidence, weak experimental evidence, and computational prediction. After obtaining the filtered results, the list of target mRNAs was manually extracted, followed by function enrichment analysis.

To explore the biological functions associated with our gene list, WebGestalt (<http://www.webgestalt.org>) was used. After uploading the gene list and selecting *Homo Sapiens* as the organism, Over-Representation Analysis (ORA) was performed, comparing our gene list to a reference background set consisting of the entire genome. Functional enrichment analysis, including Gene Ontology (GO), pathway analysis, and DO, were performed. To identify significantly enriched functional categories, WebGestalt uses Fisher's Exact Test coupled with Benjamini-Hochberg False Discovery Rate (FDR) correction to control for multiple testing. The significance threshold was set at an FDR-adjusted p-value ≤ 0.05 .

To explore the relationships between the genes and proteins, NetworkAnalyst (<https://www.networkanalyst.ca/>) was used to visualize complex biological networks. A signalling network by selecting an interaction database such as Signor. The batch selection method was applied to identify key regulatory modules by selecting highly connected and functionally hub genes within the constructed network.

Table 1. List of DE-asRNAs identified in gallbladder cancer

S. No.	Ensemble ID	Log-2-Fold Change	p value	Gene Description and symbol	Regulation
1.	ENSG00000272620	6.24	0.001	AFAP1 antisense RNA 1 (AFAP1-AS1)	Upregulation
2.	ENSG00000197301	6.07	0.002	HMGA2 antisense RNA 1 (HMGA2-AS1)	Upregulation
3.	ENSG00000174403	-7.11	0.004	MIR1-1HG antisense RNA 1 (MIR1-1HG-AS1)	Downregulation
4.	ENSG00000237125	-3.84	0.016	HAND2 antisense RNA 1 (HAND2-AS1)	Downregulation
5.	ENSG00000243479	5.95	0.018	MNX1 antisense RNA 1 (MNX1-AS1)	Upregulation
6.	ENSG00000237512	6.54	0.019	UNC5B antisense RNA 1 (UNC5B-AS1)	Upregulation
7.	ENSG00000227533	4.76	0.022	SLC2A1 antisense RNA 1 (SLC2A1-AS1)	Upregulation
8.	ENSG00000254560	5.54	0.031	BBOX1 antisense RNA 1 (BBOX1-AS1)	Upregulation
9.	ENSG00000227640	5.42	0.035	SOX21 antisense divergent transcript 1 (SOX21-AS1)	Upregulation
10.	ENSG00000277631	-5.19	0.039	PGM5P3 antisense RNA 1 (PGM5P3-AS1)	Downregulation
11.	ENSG00000234456	-3.19	0.042	MAGI2 antisense RNA 3 (MAGI2-AS3)	Downregulation
12.	ENSG00000231943	-5.26	0.044	PGM5P4 antisense RNA 1 (PGM5P4-AS1)	Downregulation
13.	ENSG00000236081	4.45	0.044	ELFN1 antisense RNA 1 (ELFN1-AS1)	Upregulation
14.	ENSG00000230061	5.21	0.044	TRPM2 antisense RNA (TRPM2-AS)	Upregulation
15.	ENSG00000267432	5.65	0.045	DNAH17 antisense RNA 1 (DNAH17-AS1)	Upregulation
16.	ENSG00000232093	5.61	0.046	DCST1 antisense RNA 1 (DCST1-AS1)	Upregulation
17.	ENSG00000261373	4.44	0.049	VPS9D1 antisense RNA 1 (VPS9D1-AS1)	Upregulation

Results

Demographic details

There were ten patients, three of whom were male and seven females. Patients with GBC had an average age of 56 years. Six GBCs were stage II, while four were stage III. All patients had adenocarcinoma. Of these, three, five, and two, respectively, were well-differentiated, moderately differentiated, and poorly differentiated. None of the patients had any family history of cancer.

Identification of DE-asRNAs

As described earlier by Dixit et al. (14), RNA sequencing generated 6,13,58,096 M total reads of raw data, including 9.878653 G total bases. 60.046748 M clean data were obtained after filtering contaminated and low-quality reads. In this study, 891 differentially expressed asRNAs (DE-asRNAs), including 468 upregulated asRNAs and 423 downregulated asRNAs, were identified. Of 891 asRNAs, 17 significant DE-asRNAs (p value <0.05) were identified and further investigated.

Twelve asRNAs, AFAP1-AS1, HMGA2-AS1, MNX1-AS1, UNC5B-AS1, SLC2A1-AS1, BBOX1-AS1, SOX21-AS1, ELFN1-AS1, TRPM2-AS, DNAH17-AS1, DCST1-AS1 and VPS9D1-AS1, were upregulated, and 5 MIR1-1HG-AS1, HAND2-AS1, PGM5P3-AS1, MAGI2-AS3, and PGM5P4-AS1, were downregulated.

PGM5P3-AS1, and MAGI2-AS3 were downregulated (Table 1). The volcano plot illustrating variations between each DE-asRNA is shown in Figure 1.

GO and pathway enrichment analysis

Out of 17, only seven asRNAs were found to be GO annotated. As a result, these 7 DE-asRNAs were significantly enriched in development, developmental process, induction of apoptosis, apoptosis, type I programmed cell death, apoptotic process, activation of apoptosis, apoptotic cell death, methylation, and cell proliferation (Table 2).

AFAP1-AS1 was significantly enriched in cell proliferation, induction of apoptosis, type I programmed cell death, apoptotic process, activation of apoptosis, apoptotic cell death, development, developmental process, cell cycle, and methylation. HAND2-AS1 was enriched in development, developmental processes, methylation, and transcription.

In addition, UNC5B-AS1, TRPM2-AS, and SOX21-AS1 were enriched in the induction of apoptosis, type I programmed cell death, apoptotic process, activation of apoptosis, and apoptotic cell death. Moreover, MNX1-AS1 was enriched in the G1 phase, RNA interference, cell cycle arrest, and cell migration.

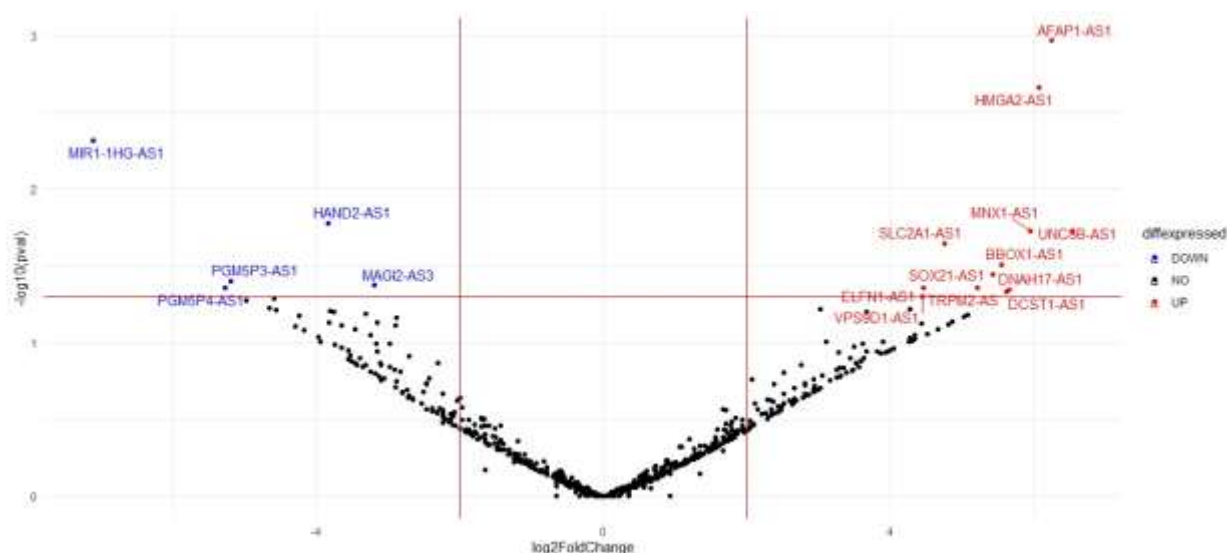


Figure 1. A volcano plot representation of significantly identified DE-asRNAs in GBC. The highlighted DE-asRNAs represent points-of-interest that display both large magnitude fold-changes (x-axis, $FC \geq 2.0$) and high statistical significance ($-\log_{10}$ of p -value, y-axis). The red colored points indicate upregulated DE-asRNAs, whereas blue colored points indicate downregulated DE-asRNAs.

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Table 2. Gene Ontology through LANTERN for the 7 DE-asRNAs

S. No.	asRNAs ID	Ontology_ID	Associated Annotation
1	AFAP1-AS1	GO:0008283	cell proliferation
		GO:0005884	actin filament
		GO:0006915	induction of apoptosis, apoptosis, type I programmed cell death, apoptotic process, activation of apoptosis, apoptotic cell death
		GO:0040007	growth, growth pattern
		GO:0009058	biosynthesis, formation, synthesis
		GO:0065007	biological regulation, regulation
		GO:0032502	development, developmental process
		GO:0016477	cell migration
		GO:0032259	methylation
		GO:0009388	antisense RNA
		GO:0007049	cell cycle
		GO:0007165	signalling pathway, signalling cascade, signal transduction, signaling cascade, signaling pathway
		GO:0007050	cell cycle arrest
		GO:0015283	bcl-2
		GO:0042060	wound healing
		GO:0070701	mucus, mucous
		GO:0003899	transcriptase
		GO:0004228	mmp-2
		GO:0006351	transcription
		GO:0016049	cellular growth, cell growth, cell expansion
2	HAND2-AS1	GO:0032502	development, developmental process
		GO:0006351	transcription
		GO:0000982	transcription factor
		GO:0032259	methylation
		GO:0003899	transcriptase
		GO:0048468	cell development, terminal differentiation
		GO:0003170	heart valve development, cardiac valve development
		GO:0007507	heart development, cardiac development
		GO:0000785	chromatin
		GO:0007568	aging, ageing
		GO:0006306	DNA methylation
3	MNX1-AS1	GO:0008283	cell proliferation
		GO:0009058	biosynthesis, formation, synthesis
		GO:0006915	induction of apoptosis, apoptosis, type I programmed cell death, apoptotic process, activation of apoptosis, apoptotic cell death
		GO:0009405	virulence, pathogenesis
		GO:0007049	cell cycle
		GO:0051318	G1 phase
		GO:0016246	RNA interference
		GO:0007050	cell cycle arrest
		GO:0016477	cell migration
		GO:0042060	wound healing
		GO:0015283	bcl-2
		GO:0016538	cyclin

4	UNC5B-AS1	GO:0006915	induction of apoptosis, apoptosis, type I programmed cell death, apoptotic process, activation of apoptosis, apoptotic cell death
		GO:0008219	cell death
		GO:0007165	signalling pathway, signalling cascade, signal transduction, signaling cascade, signaling pathway
		GO:0009405	virulence, pathogenesis
		GO:0004208	caspase-3 activity
		GO:0065007	biological regulation, regulation
5	SOX21-AS1	GO:0006954	inflammatory response, inflammation
		GO:0009058	biosynthesis, formation, synthesis
		GO:0006915	induction of apoptosis, apoptosis, type I programmed cell death, apoptotic process, activation of apoptosis, apoptotic cell death
		GO:0032259	methylation
		GO:0008283	cell proliferation
		GO:0001171	reverse transcription
		GO:0009388	antisense RNA
		GO:0006306	DNA methylation
6	ELFN1-AS1	GO:0032502	development, developmental process
		GO:0022403	cell cycle phase
7	TRPM2-AS	GO:0009388	antisense RNA
7	TRPM2-AS	GO:0006915	induction of apoptosis, apoptosis, type I programmed cell death, apoptotic process, activation of apoptosis, apoptotic cell death
		GO:0008219	cell death
		GO:0005622	intracellular
		GO:0015283	bcl-2
		GO:0005886	Juxta-membrane, cytoplasmic membrane, cellular membrane, cell membrane, plasma membrane
		GO:0032502	development, developmental process
		GO:0005576	extracellular region, extracellular
		GO:0022831	ion channels
		GO:0009058	biosynthesis, formation, synthesis
		GO:0007049	cell cycle
		GO:0023052	signalling, signaling
		GO:0007050	cell cycle arrest
		GO:0051899	membrane depolarization
		GO:0006974	DNA damage response, cellular DNA damage response, response to genotoxic stress
		GO:0045087	innate immune response, innate immunity
		GO:0007610	behaviour, behavior
		GO:0008283	cell proliferation
		GO:0005694	chromosome
		GO:0005164	tumor necrosis factor
		GO:0016020	membrane

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Table 3. Pathway enrichment analysis through ENCORI

S. No	Gene name	Associated Pathways	log10 (p-val)	log10 (FDR)
1	SLC2A1-AS1	G1 S Specific Transcription	-2.337	-1.614
		BIOCARTA CM Pathway	-2.361	-1.571
		Metabolism Of Proteins	-2.223	-1.558
		Rap1 Signalling	-2.386	-1.516
		Activation of the Pre Replicative Complex	-2.125	-1.511
		E2f Mediated Regulation of DNA Replication	-2.073	-1.505
		Association Of Licensing Factors with The Pre Replicative Complex	-2.470	-1.504
		Cdt1 Association with The Cdc6 Orc Origin Complex	-1.870	-1.343
		Synthesis Of DNA	-1.656	-1.343
		Orc1 Removal from Chromatin	-1.792	-1.338
		Ribosome	-1.675	-1.337
		Asparagine N Linked Glycosylation	-1.710	-1.318
		M G1 Transition	-1.710	-1.318
		Assembly of the Pre Replicative Complex	-1.805	-1.316
		Immune System	-1.729	-1.307
		G1 S Transition	-1.571	-1.304
2	DCST1-AS1	Spliceosome	-1.609	-1.609
3	VPS9D1-AS1	Long Term Depression	-2.469	-1.566
		Inhibition Of Insulin Secretion by Adrenaline Noradrenaline	-2.917	-2.071
		Regulation Of Insulin Secretion	-2.346	-2.065
		G Alpha S Signalling Events	-2.232	-2.023
		Neurotransmitter Receptor Binding and Downstream Transmission in The Postsynaptic Cell	-2.178	-2.002
		Integration Of Energy Metabolism	-2.235	-1.992
		ADP Signalling Through P2ry12	-2.992	-1.971
		G Alpha I Signalling Events	-2.024	-1.906
		Platelet Activation Signaling and Aggregation	-1.996	-1.905
		Transmission Across Chemical Synapses	-2.045	-1.899
		Adenylate Cyclase Inhibitory Pathway	-3.201	-1.878
		Neuronal System	-1.869	-1.802
		Hemostasis	-1.646	-1.603
		GPCR Downstream Signaling	-1.409	-1.387
		Signaling By GPCR	-1.351	-1.351
4	HAND2-AS1	RNA Degradation	-1.549	-1.373
		Small Cell Lung Cancer	-1.398	-1.319
5	BBOX1-AS1	Response To Elevated Platelet Cytosolic Ca2	-2.365	-1.939
		Lysosome	-2.232	-1.931
		Sphingolipid Metabolism	-2.476	-1.874
		Platelet Activation Signaling and Aggregation	-1.996	-1.871
		Glycosphingolipid Metabolism	-2.735	-1.832
		Phospholipid Metabolism	-2.018	-1.814
		Metabolism Of Lipids and Lipoproteins	-1.635	-1.635
		Hemostasis	-1.646	-1.588

6	UNC5B-AS1	Hedgehog Signaling Pathway	-2.266	-1.898
		Class B 2 Secretin Family Receptors	-2.070	-1.827
		Hedgehog 2pathway	-2.671	-1.826
		Basal Cell Carcinoma	-2.274	-1.730
		Pathways In Cancer	-1.501	-1.355
		GPCR Ligand Binding	-1.407	-1.340
7	SOX21-AS1	Formation Of the Hiv1 Early Elongation Complex	-2.007	-1.559
		RNA Pol Iii Transcription	-2.019	-1.542
		Influenza Life Cycle	-2.851	-1.529
		Elongation Arrest and Recovery	-2.033	-1.523
		Nucleotide Excision Repair	-1.831	-1.509
		RNA Pol Ii Transcription Pre Initiation and Promoter Opening	-1.926	-1.507
		Transcription Coupled Ner Tc Ner	-1.885	-1.493
		Formation Of RNA Pol Ii Elongation Complex	-1.885	-1.493
		mRNA Splicing Minor Pathway	-1.885	-1.493
		Formation Of Transcription Coupled Ner Tc Ner Repair Complex	-2.061	-1.479
		mRNA Capping	-2.061	-1.479
		RNA Pol Iii Transcription Initiation from Type 3 Promoter	-2.123	-1.454
		Regulatory RNA Pathways	-2.123	-1.454
		RNA Pol Ii Pre Transcription Events	-1.754	-1.453
		Formation Of the Ternary Complex and Subsequently The 43s Complex	-1.671	-1.390
		Influenza Viral RNA Transcription and Replication	-3.009	-1.386
		Activation of the mRNA Upon Binding of The Cap Binding Complex and Eifs and Subsequent Binding To 43s	-1.616	-1.355
		RNA Pol Iii Transcription Termination	-2.259	-1.334
		Abortive Elongation of Hiv1 Transcript in the Absence of Tat	-2.176	-1.331
		RNA Pol Iii Transcription Initiation from Type 2 Promoter	-2.176	-1.331
8	MNX1-AS1	MicroRNA miRNA Biogenesis	-2.176	-1.331
		DNA Repair	-1.493	-1.301
		WNT signaling	-1.766	-1.544
		Destabilization Of mRNA by Auf1 Hnmp D0	-1.990	-1.513
		Activation of the mRNA Upon Binding of the Cap Binding Complex and Eifs and Subsequent Binding To 43s	-1.791	-1.460
		Regulation Of mRNA Stability by Proteins That Bind Au Rich Elements	-1.791	-1.460
		Deadenylation Dependent mRNA Decay	-2.033	-1.459
		Eif4 Pathway	-2.333	-1.458
		Beta Catenin NUC Pathway	-1.812	-1.414
		Ps1 Pathway	-2.051	-1.352

Using the ENCORI tool, the most significantly enriched predicted pathways are shown in Table 3. HAND2-AS1 was found to be associated with small cell lung cancer and RNA degradation pathways. In addition, MNX1-AS1 was significantly enriched in WNT signaling, deadenylation-dependent mRNA decay, and the Eif4 pathway.

UNC5B-AS1 asRNA was observed to interact with pathways in cancer, basal cell carcinoma, and the hedgehog signaling pathway. Pathways associated with SLC2A1-AS1 were RAP1 signaling and E2f-mediated regulation of DNA replication. BBOX1-AS1 and VPS9D1-AS1 were associated with the hemostasis pathway. SOX21-AS1 was significantly involved in regulatory RNA pathways, the HIV-1 early elongation complex formation, RNA Pol III Transcription, and the influenza life cycle.

DO annotation

Relationships between asRNAs and other diseases were determined using Disease Ontology (DO) analysis. This study provides the first evidence of these 14 new asRNAs (UNC5B-AS1, SLC2A1-AS1, BBOX1-AS1, SOX21-AS1, ELFN1-AS1, TRPM2-AS, DNAH17-AS1, DCST1-AS1, VPS9D1-AS1, MIR1-1HG-AS1, HAND2-AS1, PGM5P4-AS1, PGM5P3-AS1, and MAGI2-AS) with GBC; the other three, i.e., AFAP1-AS1, HMGA2-AS1, and MNX1-AS1, were

previously described with GBC. However, it was discovered that 17 asRNAs were linked to various disorders, as shown in [Supplementary Table 1](#).

Functional analysis of asRNAs targeted mRNAs

A total of 12 significant asRNAs i.e., AFAP1-AS1, HMGA2-AS1, MNX1-AS1, SLC2A1-AS1, SOX21-AS1, DNAH17-AS1, VPS9D1-AS1, MIR1-1HG-AS1, HAND2-AS1, PGM5P4-AS1, TRPM2-AS, and MAGI2-AS3 were found to be correlated with the expression of mRNAs coexpressed with 577, 1, 1, 1, 5, 9, 4, 1, 8, 1, 2, and 17 mRNAs, respectively ([Supplementary Table 2](#)).

The top 24 mRNAs targeted asRNAs, including AKT1, BCL2, CASP9, SMAD2, PRPF4, ZBTB44, KRT4, EGFR, SCD, PTEN, MDM2, CCND1, MYC, DPYSL2, HIF1A, PSMB2, TAF8, MAPK14, PAFAH1B2, CDKN1C, TINCR, TRPM2, THEG, and TMEM154, were selected and extracted.

A union network of signaling and gene-miRNA networks was constructed with 1808 nodes and 2735 edges, as shown in Figure 2 (a). A significant module was constructed through the batch selection, as shown in Figure 2 (b).

To further investigate the biological functions of these targeted mRNAs, we conducted GO, Panther pathway, and DO analyses, as shown in [Supplementary](#)

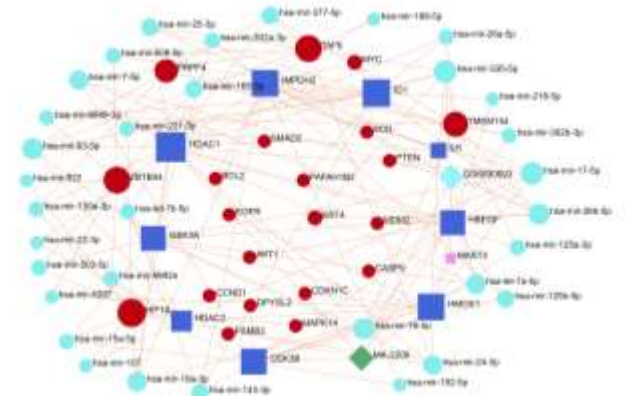
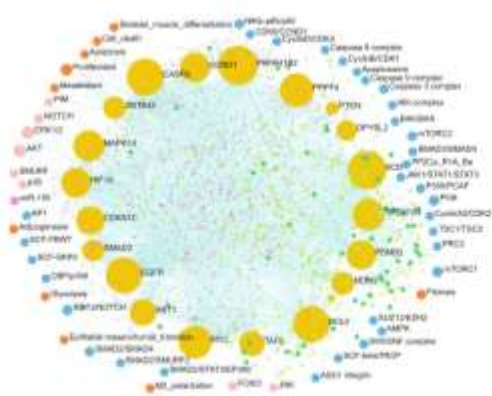


Figure 2. Left: asRNA-targeted mRNA interaction network. Nodes in the network are represented by dots with various colors and shapes. The link between the two nodes is defined by the edge. Genes (circle yellow), complex (circle light blue), miRNA (circle dark), chemical (square dark green), protein family (circle peach), small molecule (circle purple), stimulus (circle brown), phenotype (circle orange), and protein (circle pink).

Right: asRNA-targeted miRNA-interacting module. Different colored and shaped points represent nodes in the network. The edge represents the link between two nodes. Seed (circle dark red), miRNA (circle light blue), other (square dark blue), chemical (square dark green), and protein (square pink).

[Figure 1a, 1b, and 1c](#), respectively. The results indicated that for GO-BP analysis, most of the mRNAs were enriched in metabolic process, biological regulation, and developmental process. In contrast, the nucleus and membrane-enclosed lumen were enriched in GO-CC. In addition, GO-MF was enriched in protein binding and transferase activity. In addition, a high level of GO annotation showed that the majority of targeted mRNAs were enriched in regulating biological quality, regulation of the developmental process, response to stress, and cell population proliferation, as shown in [Supplementary Table 3](#). By analyzing the pathway results, we found that these mRNAs were significantly linked with the PI3K pathway, p53 pathway, apoptosis signaling pathway, and VEGF signaling pathway.

Discussion

Since the first discovery of asRNA, research has gradually advanced and made notable strides. Only a few antisense RNA functions have been shown *in vivo*, and the underlying processes underpinning these functions are still poorly understood. Recent research has shown that ncRNAs play significant and diverse roles in development and tumor progression (22, 23). It is crucial to comprehend the exact biological process by which asRNAs function to explore novel possible approaches for early detection and treatment.

In the present study, an in-depth *in silico* technique was applied to investigate significant asRNAs identified through RNA sequencing and comprehend the molecular mechanisms that contribute to the development of GBC. The study identified 17 significant asRNAs, of which 12 showed upregulation and 5 showed downregulation.

The downregulated asRNAs were significantly annotated in methylation, development, developmental process, and RNA antisense. In this study, AFAP1-AS1, HMGA2-AS1, MNX1-AS1, UNC5B-AS1, SLC2A1-AS1, BBOX1-AS1, SOX21-AS1, ELFN1-AS1, TRPM2-AS, DNAH17-AS1, DCST1-AS1 and VPS9D1-AS1 were upregulated. GO analysis revealed that the upregulated asRNAs were significantly annotated in development, developmental process, induction of apoptosis, apoptosis, type I programmed cell death, apoptotic process, activation of apoptosis, apoptotic cell death, and methylation in our study. Previous studies have shown that hypomethylation in GBC leads to the

abnormal expression of genes linked to many carcinogenic pathways, and the suppression of tumor suppressor genes occurs due to hypermethylation (24).

In our results, MIR1-1HG-AS1, HAND2-AS1, PGM5P4-AS1, PGM5P3-AS1, and MAGI2-AS3 were found to be downregulated. The downregulated asRNAs were associated with small-cell lung cancer and RNA degradation pathways. HAND2-AS1 has been linked to small-cell lung cancer and RNA degradation pathways. MNX1-AS1 was found to be notably enriched for WNT signaling, deadenylation-dependent mRNA degradation, and the Eif4 pathway. It has been reported that MNX1-AS1 is linked to a variety of oncogenic impacts in various malignancies, such as cholangiocarcinoma, colorectal cancer, and gastric cancer. Pathways in cancer, Hedgehog signaling pathway, and GPCR ligand binding were associated with UNC5B-AS1. SLC2A1-AS1 was found to be associated with RAP1 signaling and E2f-mediated DNA replication. BBOX1-AS1 and VPS9D1-AS1 were involved in the hemostasis pathway. SOX21-AS1 was linked to regulatory RNA pathways, HIV-1 early elongation complex formation, RNA Pol III Transcription, and the influenza life cycle.

Previous studies suggest that HAND2-AS1 (25), TRPM2-AS (26), SOX21-AS1 (27), and UNC5B-AS1 (28) are linked with PI3K/AKT signaling pathways. An active PI3K-Akt pathway modulates the activity of several substrates involved in proliferation, cell survival, and cell cycle progression; thus, this pathway is essential for cancer development (29), and a clinical trial of mTOR inhibitors is presently underway. In addition, several research studies have reported that AFAP1-AS1 (30), BBOX1-AS1 (31), and VPS9D1-AS1 (32) are associated with the Wnt/ β -catenin signaling pathway in various diseases. However, earlier studies reported that the Wnt/ β -catenin pathway is involved in embryonic and organ development and cancer growth (33). Furthermore, previous studies have suggested that UNC5B-AS1 (34) and DCST1-AS1 (35) are linked with AKT/mTOR signaling pathways. It has also been demonstrated that SOX21-AS1 functions as an oncogene and is involved in Wnt-catenin pathways in glioma cells via the SOX21-AS1/miR-144-3p/PAK7 axis (36).

The DO analysis results demonstrated that 14 novel asRNAs (UNC5B-AS1, SLC2A1-AS1, BBOX1-AS1,

SOX21-AS1, ELFN1-AS1, TRPM2-AS, DNA H17-AS1, DCST1-AS1, VPS9D1-AS1, MIR1-1HG-AS1, HAND2-AS1, PGM5P4-AS1, PGM5P3-AS1, and MAGI2-AS) were associated with GBC. The other three asRNAs (AFAP1-AS1 (37), HMGA2-AS1 (38), and MNX1-AS1(39)) have been reported in gallbladder cancer. AFAP1-AS1 has also been linked with lung cancer (40), breast cancer (41), nasopharyngeal carcinoma (42), and pancreatic ductal adenocarcinoma (43), and HMGA2-AS1 has been linked with laryngeal squamous cell carcinoma (44), osteosarcoma (45), and pancreatic cancer (46). Previous studies reported the association of MNX1-AS1 with ovarian (47), lung (48), and colorectal cancer (49).

The function of asRNAs via asRNA-targeted mRNA interactions in controlling gene expression is one of the least understood aspects of biology (50, 51, 52). A number of studies have been conducted on the association between GBC and abnormal asRNA expression, but none have reached a conclusive result. In the present study, the top 24 enriched asRNA-targeted mRNAs were identified, providing a novel aspect to understanding the gene function that is crucial for tumor development.

Interaction between asRNAs by overlapping mRNAs revealed a unique mechanism of gene regulation that plays a crucial role in cancer development. As a result, AFAP1-AS1 was found to be upregulated in GBC patients and coexpressed with the highest number of mRNAs, which were significantly enriched in cell proliferation and signaling cascades, which could be due to the maximum work being done on this asRNA.

The functional relevance of asRNAs targeting specific mRNAs has been implicated in breast carcinoma, adenocarcinoma, uterine cervical, prostatic, lung, colonic, and mammary neoplasms. AKT1 has been linked with breast cancer (53), prostate cancer (54), and colorectal cancer (55). Similarly, MYC and MAPK were also associated with breast (56, 57), prostate (58, 59), gallbladder (4), and colorectal cancer (60, 61).

Even with the significant findings being reported, it is essential to acknowledge the inherent limitations in our research. The study compared only one dataset of normal gallbladder to find the DE-asRNA, and we are comparing it with other datasets of normal gallbladder to see if the results are consistent. As the functions of

the identified asRNAs are extrapolated using bioinformatics, the information is limited. Not much work has been done on these asRNAs, and only three have been identified and studied earlier. It is assumed that after publication of these results, more work on 14 novel asRNAs will be carried out, and our understanding will improve with time. Experiments have not yet validated the findings of this study, and this too is a significant drawback. Still, we hope that researchers will pick these data up and will explore them further experimentally.

Recent research has shed light on the role of asRNAs in various cancers, including GBC. While our study provides valuable initial insights, it is essential to acknowledge certain limitations. The small sample size may reduce our analyses' statistical strength and limit our findings' broader relevance. Furthermore, the lack of experimental validation means that our conclusions are based primarily on bioinformatics predictions, even though we ensured that only those interactions that have at least one experimentally proven interaction are chosen. The future research will address these limitations by expanding the sample size and incorporating experimental validation to deepen our understanding of asRNAs in GBC.

Conclusion

The present study reports 17 DE-asRNAs in gallbladder cancer, of which 14 have never been reported before. The 3 asRNAs that have been studied interact with AKT1, MYC, and MAPK, and through this interaction, they control cell division and the cell cycle. This suggests that asRNAs may act as potential targets for GBC treatment and may play a significant role in silencing genes. However, further studies are needed to validate these findings.

Ethics approval

Ethics Statement: Ethics approval and consent to participate. The research was approved by the Institute Ethics committee (No. Dean/2018/ EC/561), and written informed consent was obtained from all patients.

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Conflict of interests

The authors declare that they have no competing interests.

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